



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO DURAI FUEL SERVICES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

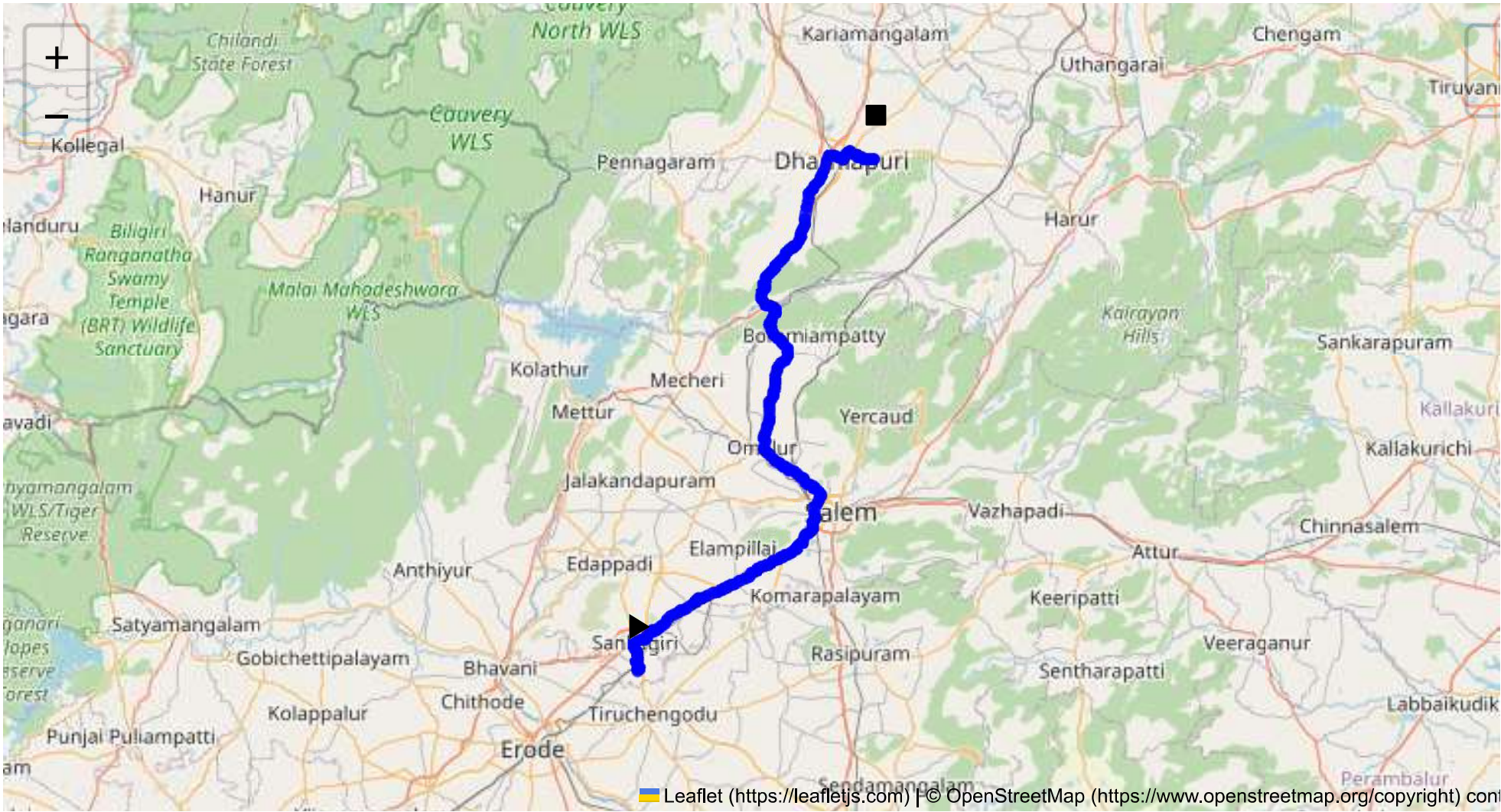
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 113.42 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.8 hours
Start: (11.4381, 77.8734)
End: (12.137043, 78.203133)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route spans approximately 113.42 kilometers, starting from CVQF+23W, Sangagiri, Tamil Nadu, to 46P3+R7, Pananthoppu, Tamil Nadu. It involves major highways and some rural roads, traversing through

diverse terrains ranging from urban areas to agricultural and forested regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Typical Weather:** The region typically experiences tropical weather with hot summers, a monsoon season from June to September, and mild winters.
- **Potential Hazards:** During monsoon, roads can become slippery and prone to flooding, especially in low-lying areas, increasing the risk of hydroplaning incidents and delays.

3. Traffic Patterns

- **Peak Hours:** Typically, traffic is heavier during morning (7 AM - 9 AM) and evening (5 PM - 7 PM) rush hours, particularly near larger towns and cities.
- **Congestion-Prone Areas:** Major intersections within municipalities and highway junctions are prone to congestion. Watch out near larger towns like Erode or Salem.

4. Assessment of Road Quality and Infrastructure

- **Highways:** Generally good, suitable for heavy vehicles.
- **Rural Roads:** Some sections may be narrower with fewer passing lanes and less-maintained surfaces. Potholes and uneven patches can be expected.

5. Suggestions for Alternative Routes

- **For Emergencies:** Use NH44 as an alternate route, though it could add time due to potential detours.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Specific routes may be designated for hazardous materials; compliance with local state transport regulations is required.
- Restrictions could be imposed during festivals or public events leading to route diversions.

7. Overview of Historical Incidents

- **Heavy Vehicles:** Some incidents recorded involve tipping or jackknifing during monsoon.
- **Hazardous Materials:** No major publicized events within the past few years, suggesting relative safety if regulations are followed.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Stay clear of any protected wildlife zones and adhere to noise and pollution regulations when near sensitive zones.

9. Analysis of Communication Coverage

- **Coverage:** Generally good along major highways, but potential dead zones may exist in deeply rural or forested regions.
- **Advice:** Carry a backup communication device with satellite connectivity in less than optimal coverage areas.

10. Estimated Emergency Response Times

- **Urban Areas:** 30-45 minutes due to proximity to emergency services.
- **Rural Areas:** Can stretch to 60-90 minutes depending on accessibility and road conditions.

11. Overall Summary of Risk Assessment

- **Overall Risk:** Moderate for heavy vehicles carrying hazardous materials. The main concerns are weather impacts and road maintenance in certain areas.
- **Recommendations:** Strict adherence to traffic regulations and active monitoring of weather forecasts is advised. Plan travel around peak times if possible and ensure collaboration with local authorities for smooth transportation of hazardous materials.

By taking the above factors into consideration, truck drivers can minimize risks and ensure a safer journey from Sangagiri to Pananthoppu.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.13658, 78.16171	15 KM/Hr
1	Roundabout	High	12.14589, 78.17156	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr
8	Turn	Medium	11.46400, 77.87323	30 KM/Hr
9	Turn	Medium	11.46765, 77.87037	30 KM/Hr
10	Turn	High	11.47403, 77.86841	15 KM/Hr
11	Turn	High	11.49149, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.49224, 77.89601	15 KM/Hr
13	Turn	Medium	11.53512, 77.95717	30 KM/Hr
14	Turn	Medium	11.53606, 77.95869	30 KM/Hr
15	Turn	Medium	11.64183, 78.11810	30 KM/Hr
16	Turn	Medium	11.64338, 78.11998	30 KM/Hr
17	Turn	Medium	11.92556, 78.06499	30 KM/Hr
18	Turn	Medium	11.95854, 78.04990	30 KM/Hr
19	Turn	Medium	11.96015, 78.04945	30 KM/Hr
20	Turn	Medium	11.96370, 78.05142	30 KM/Hr
21	Turn	Medium	11.96433, 78.05113	30 KM/Hr
22	Turn	High	11.96542, 78.04938	15 KM/Hr
23	Turn	High	11.96650, 78.04927	15 KM/Hr
24	Turn	Medium	11.96804, 78.05080	30 KM/Hr
25	Turn	High	12.14045, 78.13962	15 KM/Hr
26	Turn	High	12.14497, 78.16507	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
33	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
34	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
36	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
39	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
41	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
42	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
43	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
45	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
46	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
47	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
48	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
49	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
50	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
51	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
52	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
53	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
54	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
55	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
56	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
57	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
58	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
31	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
32	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
35	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
38	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
40	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
44	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots